Problem 9.12

Interest on certain deposits is compounded monthly. Find the nominal rate of interest, per annum, for these deposits if the APR is 15%. Obtain the accumulation of an investment of $1,000 over a period of 6 months. How much interest has accrued by that time?

Solution

Let $i^{(12)}$ denote the nominal annual rate of interest compounded monthly. Then the monthly rate is

$$\frac{i^{(12)}}{12}.$$

Since the APR is the annual effective rate, we have

$$1+i=\left(1+\frac{i^{(12)}}{12}\right)^{12},$$

where

$$i=0.15.$$

Therefore,

$$\begin{aligned}1.15&=\left(1+\frac{i^{(12)}}{12}\right)^{12},\\ 1.15^{1/12}&=1+\frac{i^{(12)}}{12},\\ i^{(12)}&=12\left(1.15^{1/12}-1\right).\end{aligned}$$

Thus,

$$i^{(12)}\approx 0.140579003.$$

So the nominal annual rate of interest compounded monthly is

$$\boxed{14.0579\%}.$$

For an investment of $1,000 over 6 months, the accumulated value is

$$1000\left(1+\frac{0.140579003}{12}\right)^{6}.$$

Equivalently, since 6 months is one half of a year,

$$1000(1.15)^{1/2}=1072.380529.$$

Hence the accumulated value is approximately

$$\boxed{\$1,072.38}.$$

The interest accrued is

$$1072.380529 - 1000 = 72.380529.$$

Therefore, the interest accrued after 6 months is approximately

$$\boxed{\$72.38}.$$